Assignment 2

PCA and eigendecomposition

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Imports

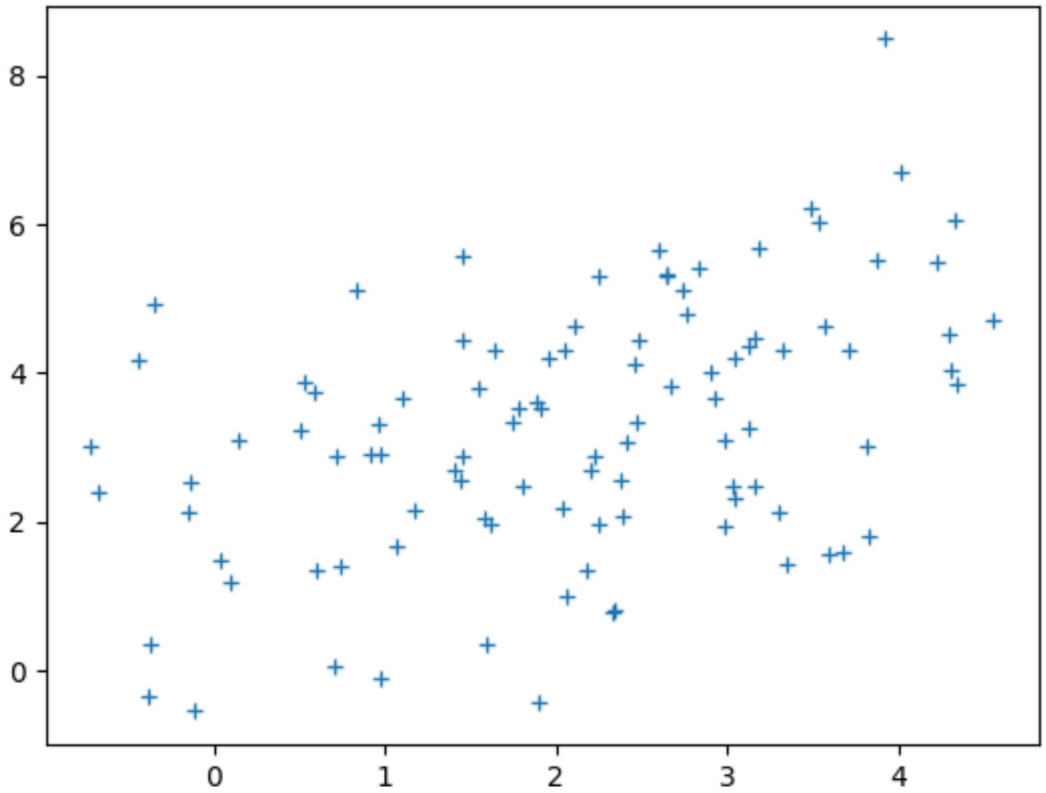
In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sympy import Matrix
```

Generate bi-variate normal random numbers

In [2]:
```
mu = [2, 3]
cov = [[2, 1.5], [1.5, 4]]

np.random.seed(42)
R = np.random.multivariate_normal(mu, cov, 100)

plt.plot(R[:, 0], R[:, 1], "+")
plt.show()
```



Check variance and correlation

In [3]:
```
sample_cov = np.cov(R, rowvar=False)
cor = np.corrcoef(R, rowvar=False)

print("Sample covariance:\n", sample_cov)
print("\nCorrelation:\n", cor)
```

```
Sample covariance:
 [[1.76781403 1.00496726]
 [1.00496726 2.94842117]]

Correlation:
 [[1.         0.44018867]
 [0.44018867 1.        ]]
```

Double check corr

In [4]:
```
print(1.00496726 / (1.76781403**0.5 * 2.94842117**0.5))
```

```
0.4401886749018094
```

Perform PCA on the bi-variate normal random numbers

In [5]:
```
pca = PCA()
pca.fit(R)

loading_vector = pca.components_.T

latent = pca.explained_variance_

score = pca.transform(R)
```

```
variability_explained = pca.explained_variance_ratio_ * 100

print("Loading vectors:\n", loading_vector)
print("\nVariance explained:\n", variability_explained)
```

```
Loading vectors:
 [[ 0.4967515   0.86789282]
 [ 0.86789282 -0.4967515 ]]

Variance explained:
 [74.71275573 25.28724427]
```

Check correlation

In [6]:
```
print(np.corrcoef(score, rowvar=False))
```

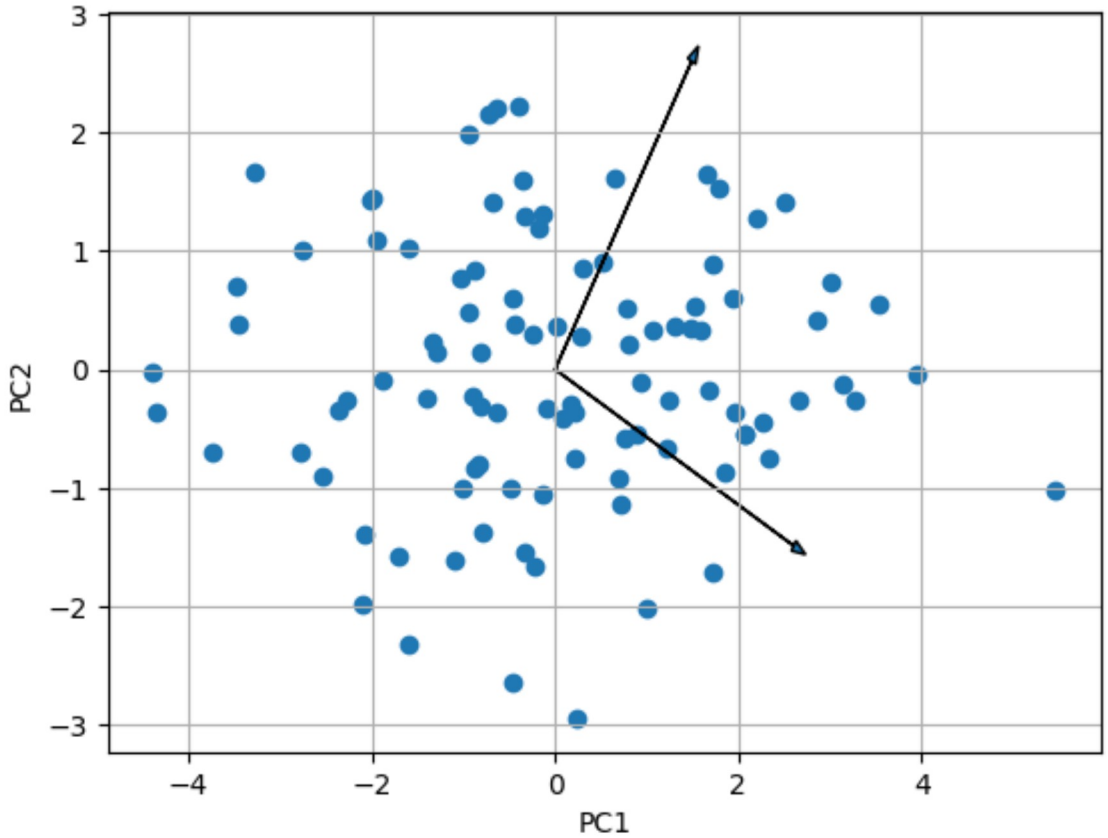
```
[[1.00000000e+00 6.04528708e-16]
 [6.04528708e-16 1.00000000e+00]]
```

In [7]:
```
print("PC1:", np.linalg.norm(loading_vector[:, 0]))
print("PC2:", np.linalg.norm(loading_vector[:, 1]))
print("Dot prod:", np.dot(loading_vector[:, 0], loading_vector[:, 1]))
```

```
PC1: 1.0
PC2: 1.0
Dot prod: 0.0
```

In [8]:
```
plt.scatter(score[:, 0], score[:, 1])
plt.arrow(0, 0, loading_vector[0, 0] * 3, loading_vector[0, 1] * 3, head_width=0
plt.arrow(0, 0, loading_vector[1, 0] * 3, loading_vector[1, 1] * 3, head_width=0

plt.xlabel("PC1")
plt.ylabel("PC2")
plt.grid(True)
plt.show()
```



Perform PCA step-by-step

In [ ]:
```
data = R
data[:, 0] = (R[:, 0] - np.mean(R[:, 0])) / np.sqrt(2.0479)
data[:, 1] = (R[:, 1] - np.mean(R[:, 1])) / np.sqrt(4.2666)

cov = np.cov(data, rowvar=False)

print(cov)
```

```
[[0.8632326  0.33998251]
 [0.33998251 0.69104701]]
```

In [ ]:
```
eigenvals, eigenvectors = np.linalg.eig(cov)
print("eigenvectors\n", eigenvectors)
print("eigenvalues\n", eigenvals)
```

```
eigenvectors
 [[ 0.78913841 -0.61421541]
 [ 0.61421541  0.78913841]]
eigenvalues
 [1.12785347 0.42642614]
```

Check eigenvalues

In [ ]:
```
print(cov @ eigenvectors[:, 0] - eigenvals[0] * eigenvectors[:, 0])
print(cov @ eigenvectors[:, 1] - eigenvals[1] * eigenvectors[:, 1])
```

```
[0. 0.]
[ 5.55111512e-17 -5.55111512e-17]
```

In [12]:
```
print(np.linalg.norm(eigenvectors[:, 0]))
print(np.linalg.norm(eigenvectors[:, 1]))
```

```
print(np.dot(eigenvectors[:, 0], eigenvectors[:, 1]))
```

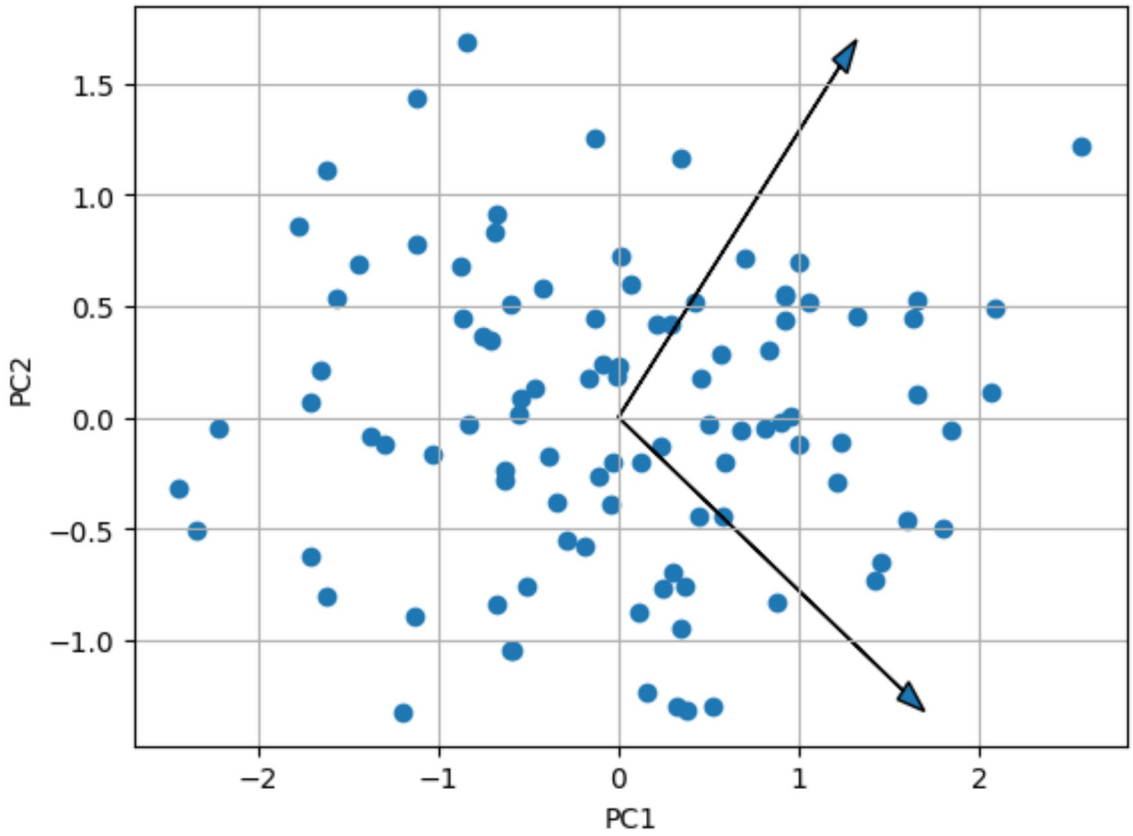
```
1.0
1.0
0.0
```

In [13]:
```
score = data @ eigenvectors
print(np.corrcoef(score, rowvar=False))
```

```
[[ 1.00000000e+00 -1.83805379e-16]
 [-1.83805379e-16  1.00000000e+00]]
```

In [14]:
```
plt.scatter(score[:, 0], score[:, 1])
plt.arrow(0, 0, eigenvectors[0, 0] * 2, eigenvectors[0, 1] * 2, head_width=0.1)
plt.arrow(0, 0, eigenvectors[1, 0] * 2, eigenvectors[1, 1] * 2, head_width=0.1)

plt.xlabel("PC1")
plt.ylabel("PC2")
plt.grid(True)
plt.show()
```



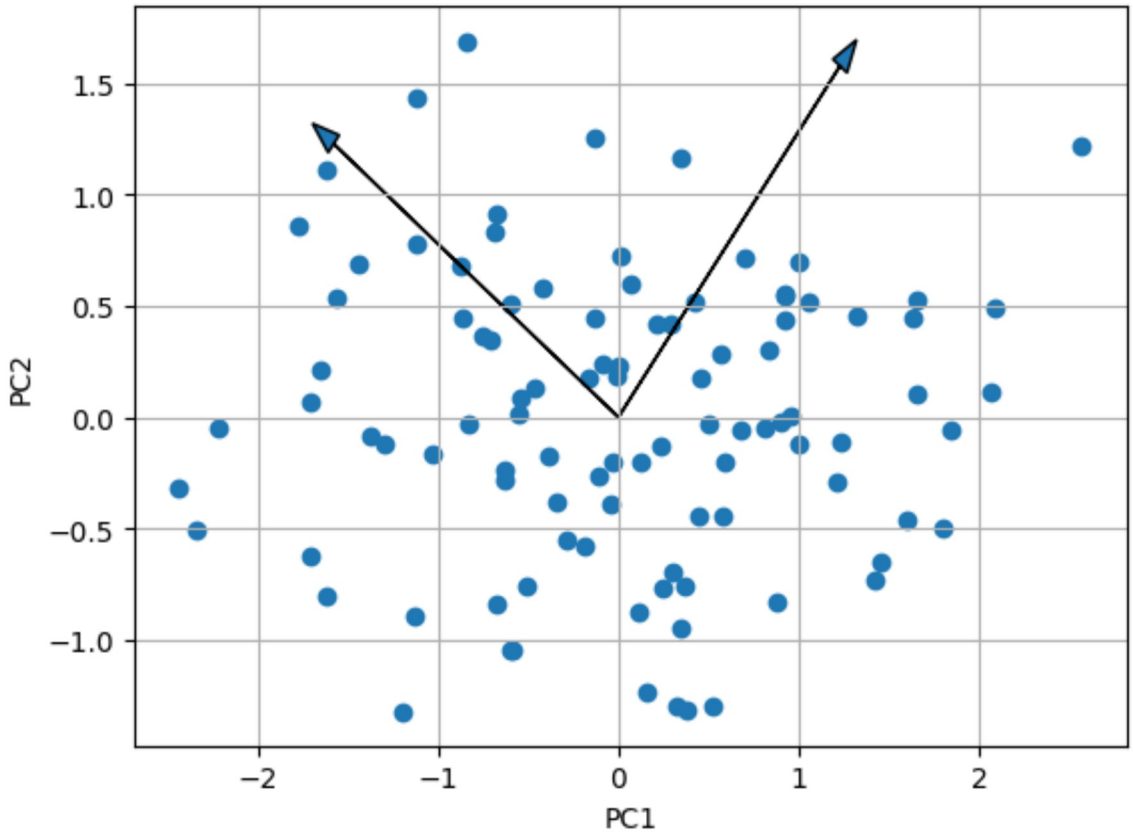
In [15]:
```
variances = np.var(score, axis=0)
print("Variances:", variances)
print("Explained variance ratio:", variances / sum(variances))
```

```
Variances: [1.11657493 0.42216188]
Explained variance ratio: [0.72564387 0.27435613]
```

Eigenvectors can be negated, probably why my arrows are pointing right while they are pointing up in the matlab example, below is the fixed version

In [19]:
```
plt.scatter(score[:, 0], score[:, 1])
plt.arrow(0, 0, -eigenvectors[0, 0] * 2, -eigenvectors[0, 1] * 2, head_width=0.1
plt.arrow(0, 0, eigenvectors[1, 0] * 2, eigenvectors[1, 1] * 2, head_width=0.1)

plt.xlabel("PC1")
plt.ylabel("PC2")
plt.grid(True)
plt.show()
```



In [ ]:
```
print("Covariance matrix\n", cov)
a = 1
b = -cov[0, 0] - cov[1, 1]
c = cov[0, 0] * cov[1, 1] - cov[0, 1] * cov[1, 0]

eigenvalues = np.roots([a, b, c])
print("Eigenvalues\n", eigenvalues)
```

```
Covariance matrix
 [[0.8632326  0.33998251]
 [0.33998251 0.69104701]]
Eigenvalues
 [1.12785347 0.42642614]
```

In [ ]:
```
lambda1 = np.diag([eigenvalues[0], eigenvalues[0]])
print("lambda 1\n", lambda1)
char1 = cov - lambda1
print(np.linalg.det(char1))

lambda2 = np.diag([eigenvalues[1], eigenvalues[1]])
char2 = cov - lambda2
print(np.linalg.det(char2))
```

```
lambda 1
 [[1.12785347 0.        ]
 [0.         1.12785347]]
1.6985538414009587e-16
4.8495258750783945e-17
```

In [ ]:
```
A1 = cov - lambda1

print("A1:\n", A1)
print("Rank A1:", np.linalg.matrix_rank(A1))
print("Det A1:", np.linalg.det(A1))
rref = np.array(Matrix(np.round(A1, 5)).rref()[0]).astype(np.float64)
print("RREF A1:\n", rref)
```

```
A1:
 [[-0.26462087  0.33998251]
 [ 0.33998251 -0.43680646]]
Rank A1: 1
Det A1: 1.6985538414009587e-16
RREF A1:
 [[1. 0.]
 [0. 1.]]
```

In [23]:
```
v1 = np.array([-rref[0, 1], 1])
print("Eigenvector 1:", v1 / np.linalg.norm(v1))
```

```
Eigenvector 1: [-0.  1.]
```

In [ ]:
```
A2 = cov - lambda2
rref2 = np.array(Matrix(np.round(A2, 5)).rref()[0]).astype(np.float64)
v2 = np.array([-rref2[0, 1], 1])
print("Eigenvector 2:", v2 / np.linalg.norm(v2))
```

```
Eigenvector 2: [-0.  1.]
```